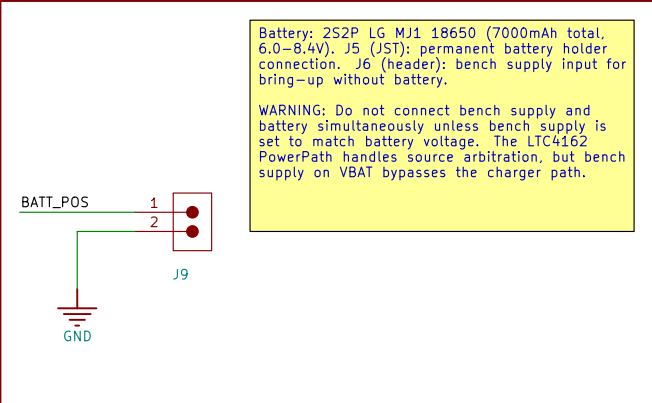


CONNECTORS & TELEMETRY

CONNECTORS & TELEMETRY – Battery interface, CSKB H1 + H2 stack connectors for inter-board power and signal distribution, status LEDs, and test points for bench validation. All subsystem power and I2C telemetry routes through the CSKB H1/H2 connectors.

Battery Connector



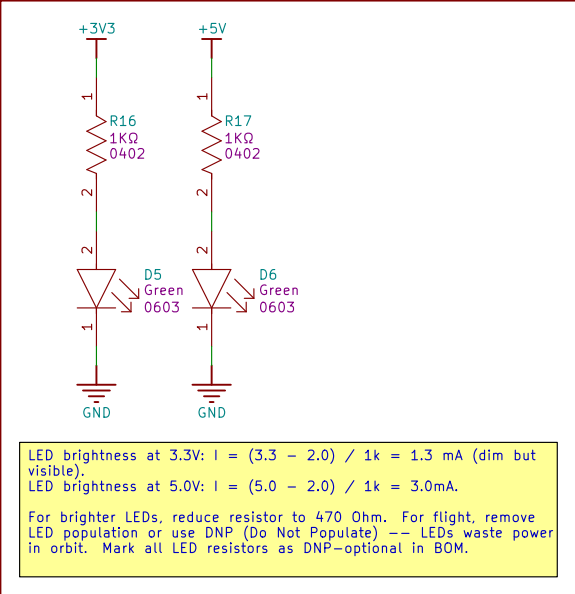
Regulation

File: Regulation.kicad_sch

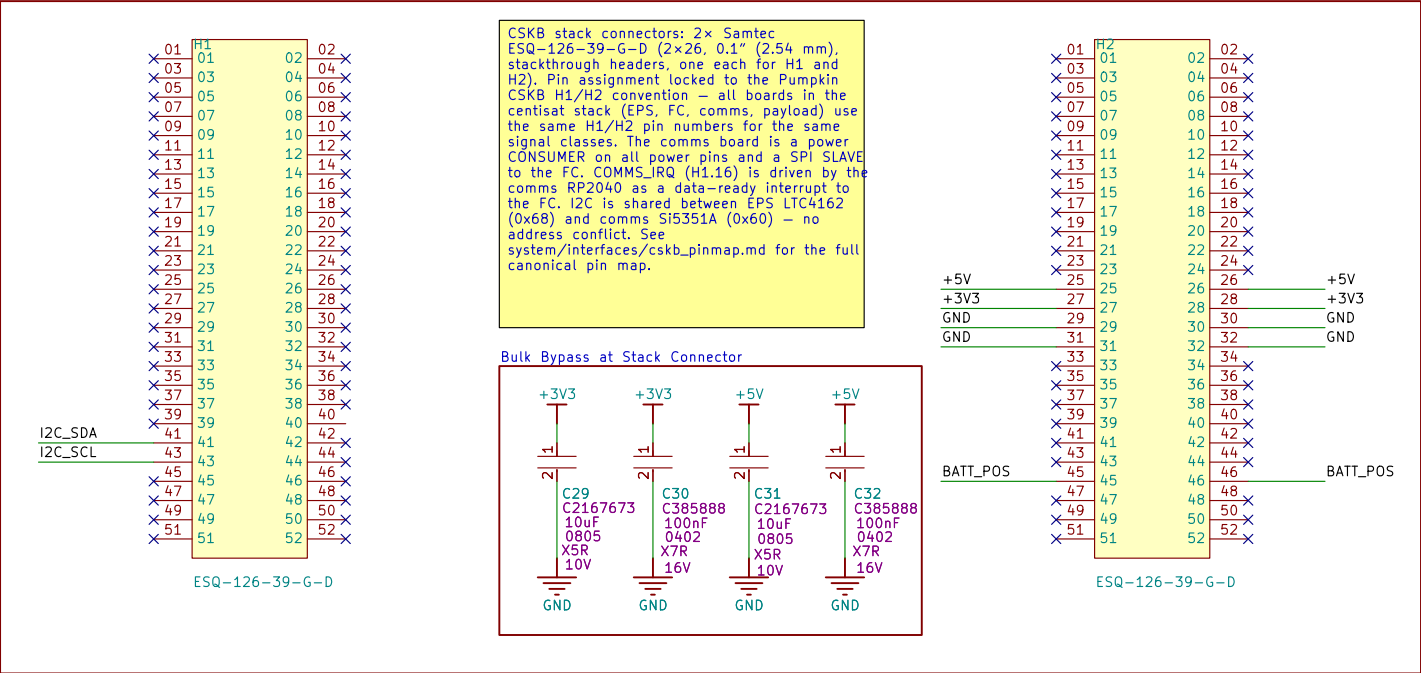
Solar_Charger

File: Solar_Charger.kicad_sch

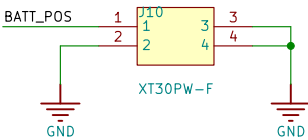
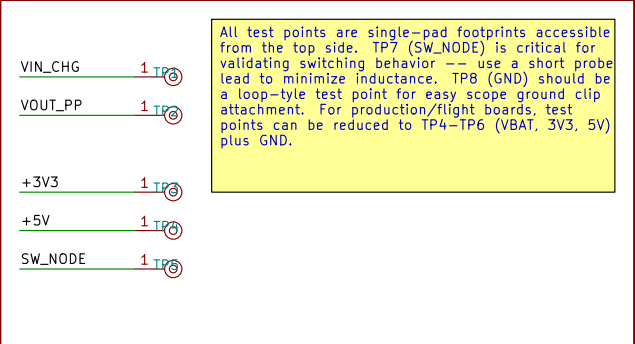
Status LEDs



CSKB H1 + H2 Stack Connectors



Test Points



Sheet: /

File: eps.kicad_sch

Title: CubeSat EPS -- Connectors & Telemetry

Size: A3

Date:

Rev: B

KiCad E.D.A. 10.0.0

Id: 1/3

REGULATION

REGULATION -- Both rails are produced by TPS62933F 3A synchronous buck converters fed from V_OUT_PP (LTC4162 PowerPath output, 6–18V). The TPS62933F has a 3.8–30V input range, which comfortably accepts the full PowerPath swing from battery-only (~6V) to peak solar charging (~8V), perserving proper PowerPath topology.

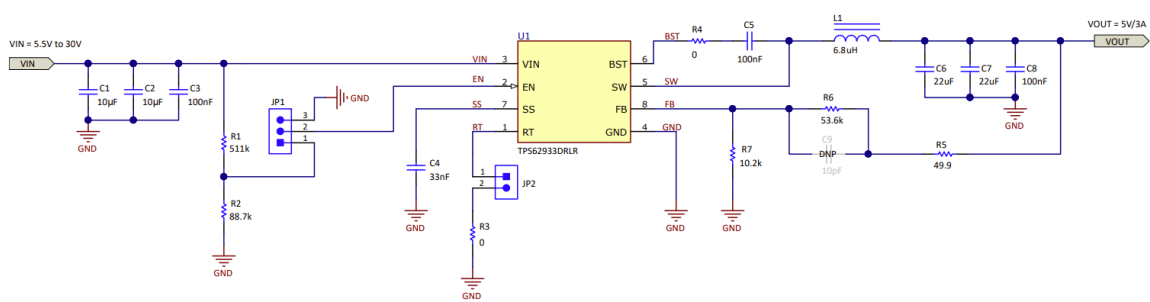
Why TPS62933F (forced CCM variant) over the standard TPS62933: – Fixed swingtching frequency at all loads (no PFM transitions) – Predictable EMI spectrum -- important since the comms board peer is RF-sensitive at 145 MHz and 437 MHz – Lower output ripple under light load – Cost: ~30% the price of TPSM5D1806, no LDO needed for 5V rail

Common switching frequency choice: 500 kHz, set by floating the RT pin (datasheet Table 9–1). Rationale: 500 kHz is the lowest standard setting and gives the best efficiency, lowest switching losses, and the cleanest fundamental away from any sensitive basebands. The two converters are not synchronized -- the RP2040 GPIOs on the flight-controller and comms boards filter their own supply locally, so beat tones between the two buck stages are not a concern.

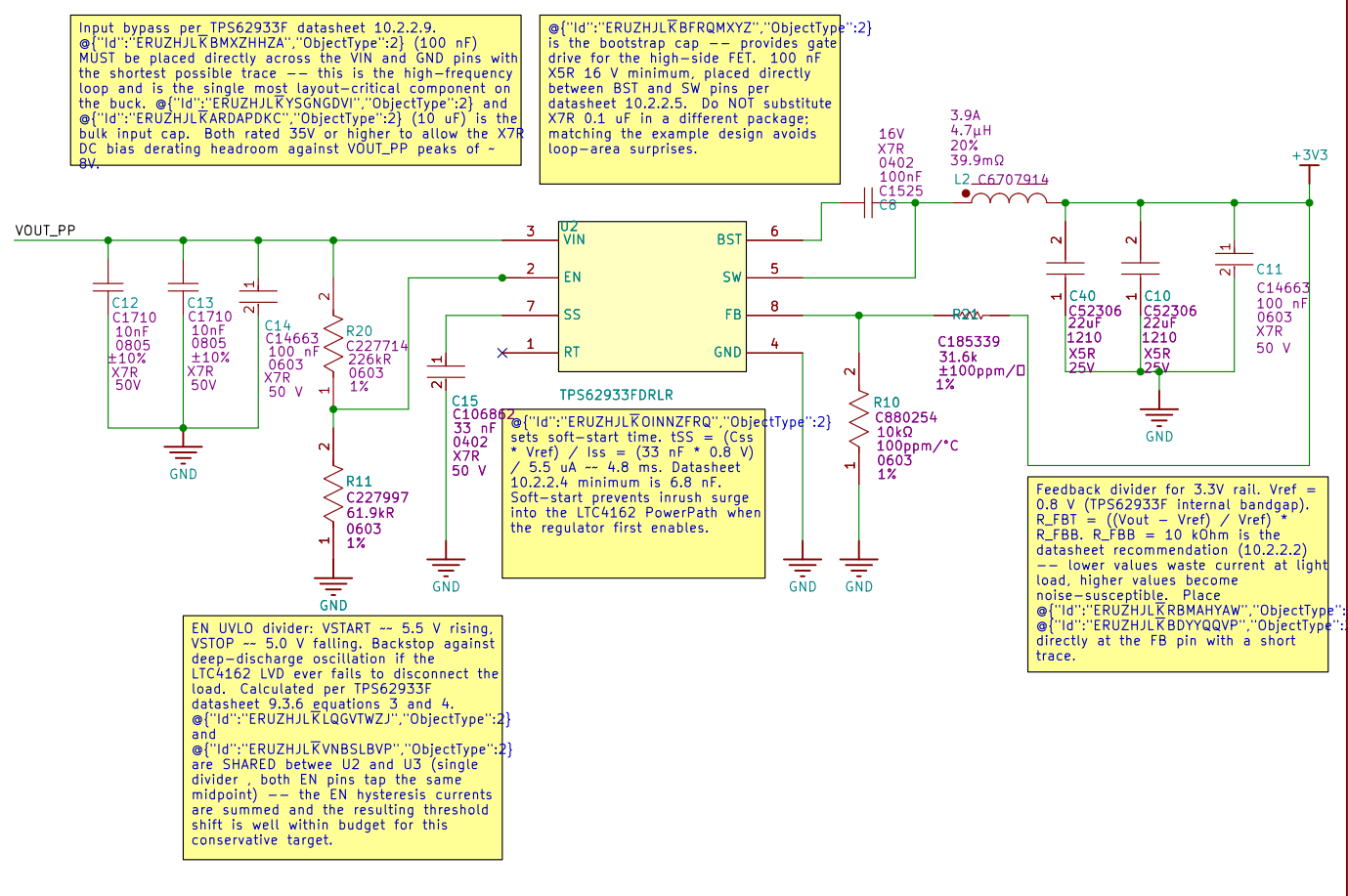
Common feedback equation: $R_{FBT} = ((V_{OUT} - 0.8) / 0.8) * R_{XFBB}$, with $R_{FBB} = 10k\Omega$ recommended (datasheet Equation 1, 10.2.2.2).

Calculation (datasheet 10.2.2.2): $R_{FBT} = ((3.3 - 0.8) / 0.8) * 10\text{ K}\Omega = 31.25\text{ k}\Omega$

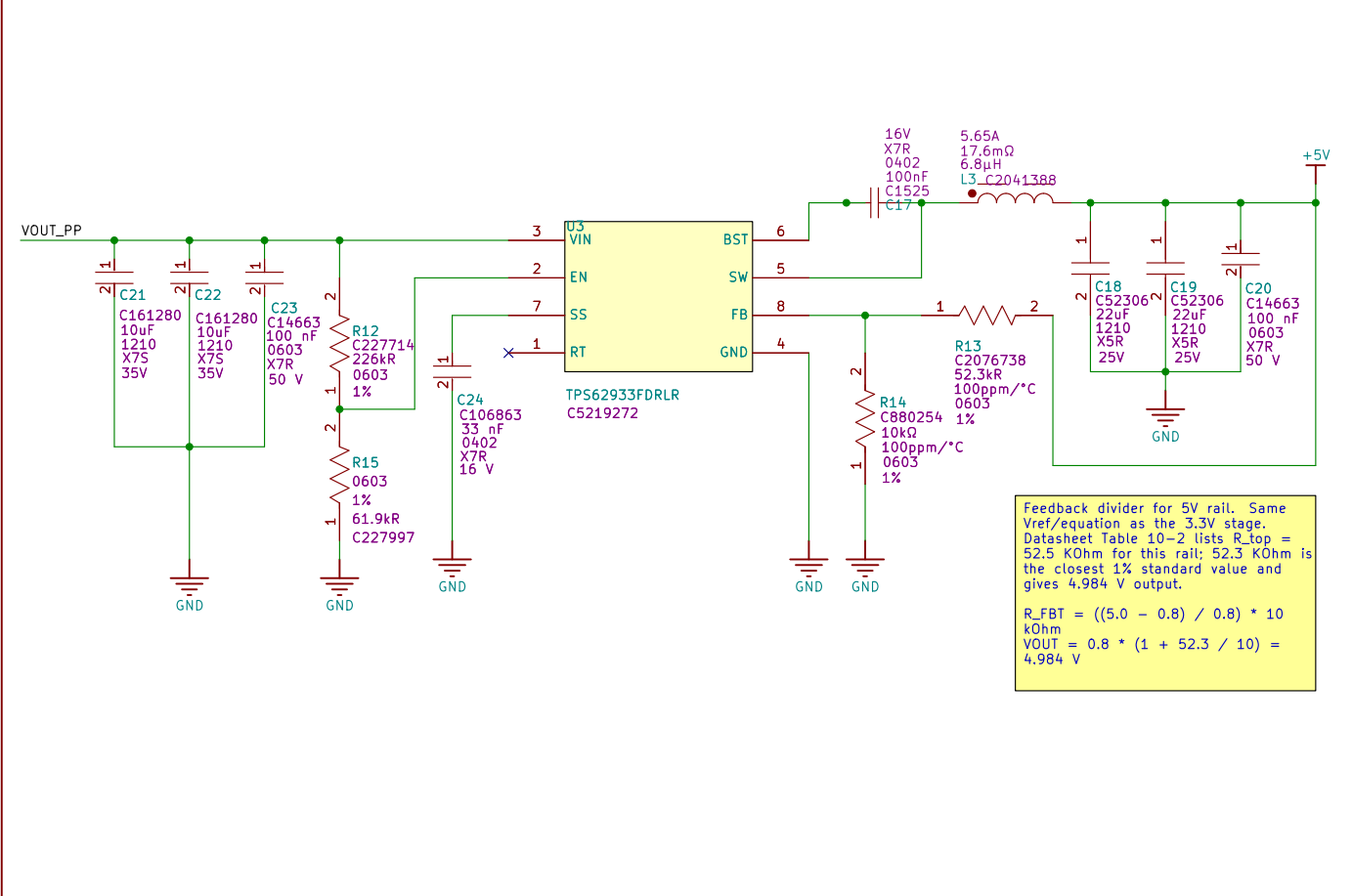
Figure 10-1. TPS62933 5-V Output, 3-A Reference Design



TPS62933F -- 3.3V Rail



TPS62933F -- 5V Rail



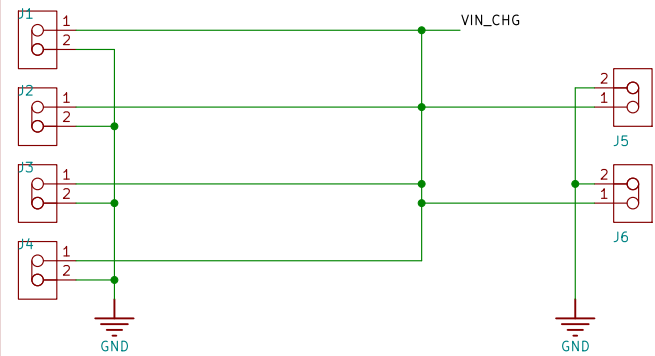
PGOOD: the TPS62933F variant (SS pin) does not provide a PGOOD output. If a power-good telemetry signal is required for the flight controller, swap one or both stages to TPS62933P (PG pin) and route an open-drain PGOOD net to Sheet 3. For the prototype, the LTC4162's I2C telemetry already exposes input/charger health and is the primary monitoring path.

SOLAR INPUT & CHARGER

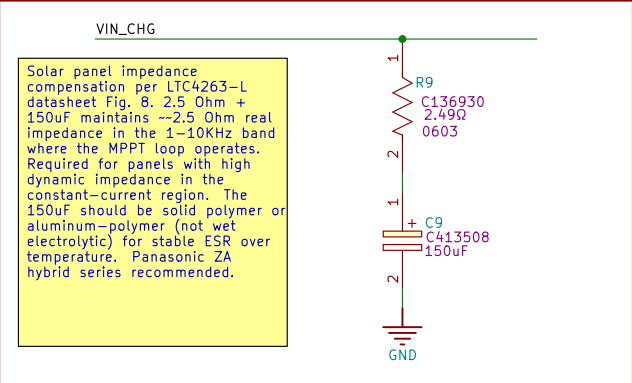
@{"Id":"NFZYWHMEPGWUXVN","ObjectType":2} Input reverse-blocking MOSFET. LTC4162 gives gate via INFET charge pump. FDMC8327L selected per DC2038A reference design. @{"Id":"NFZYWHMETBRIUZO","ObjectType":2} Input current sense. $I_{IN(MAX)} = 32nV / 10m\Omega = 3.2A$. Sized for bench supply flexibility -- actual solar panels deliver only --80mA. Use Kelvin connection (4-pad footprint preferred, sense traces from inner pads to CLP/CLN pins). See LTC4162-L datasheet Figure 5.

Solar Input

SOLAR INPUT -- 4 face PCBs (X+, X-, Y+, Y-), each with 4x SM141K10TF in series (~8.4V Vmp, ~23.6V Voc, ~41mA Imp per face). Faces are paralleled through blocking diodes on this board. Realistic orbit current: 40-80mA depending on attitude. Bench supply can simulate any condition through any single JST connector.



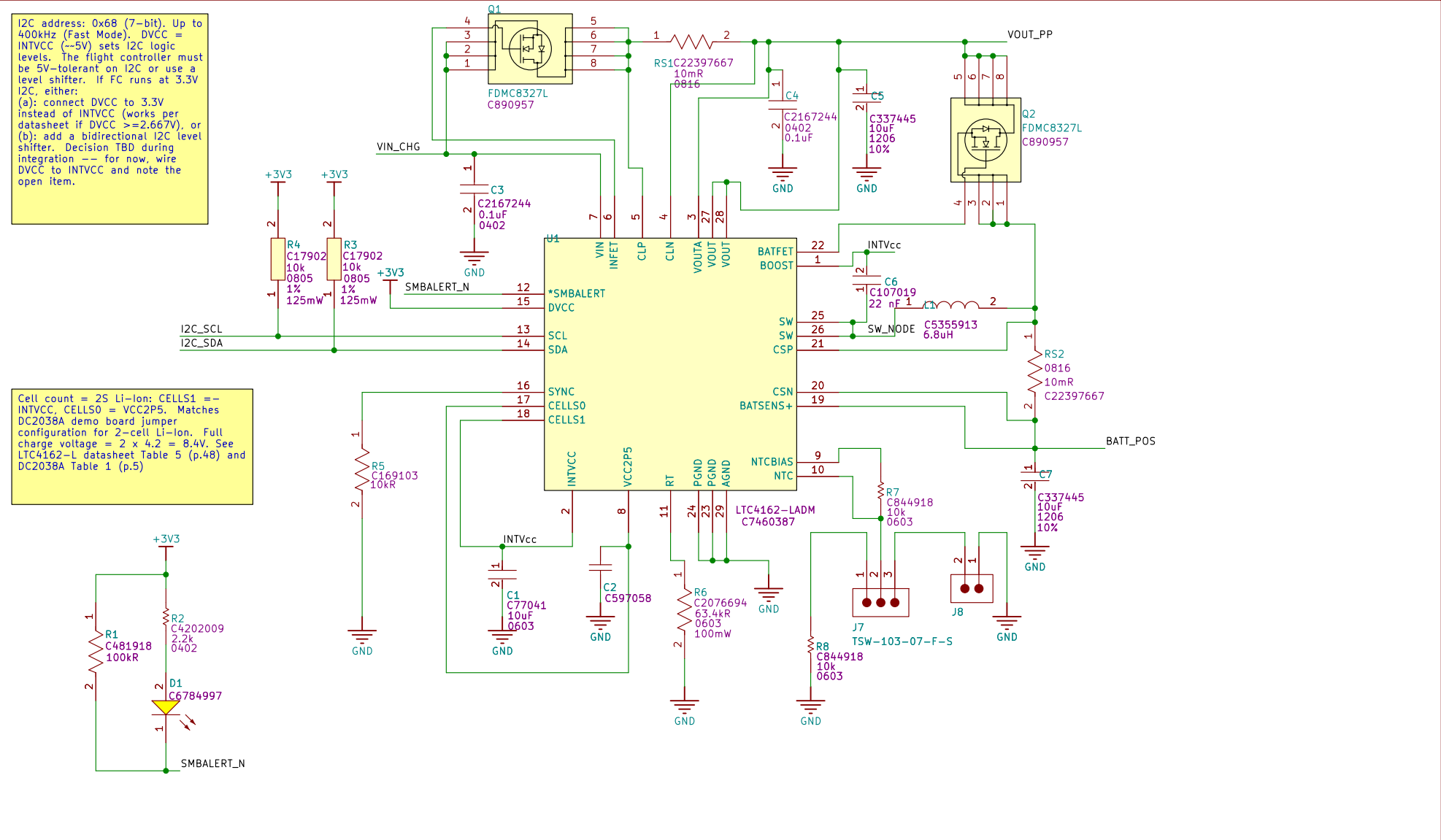
Input Impedance Compensation



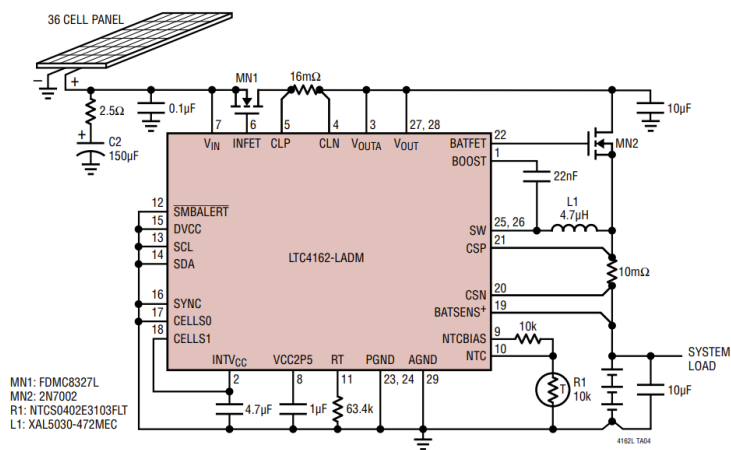
LTC4162 and PowerPath

I2C address: 0x68 (7-bit). Up to 400kHz (Fast Mode). DVCC = INTVCC (~5V) sets I2C logic levels. The flight controller must be 5V-tolerant on I2C or use a level shifter. If FC runs at 3.3V I2C, either:
(a): connect DVCC to 3.3V instead of INTVCC (works per datasheet if DVCC >= 2.667V), or
(b): add a bidirectional I2C level shifter. Decision TBD during integration -- for now, wire DVCC to INTVCC and note the open item.

Cell count = 2S Li-Ion: CELLS1 = INTVCC, CELLS0 = VCC2P5. Matches DC2038A demo board jumper configuration for 2-cell Li-Ion. Full charge voltage = 2 x 4.2 = 8.4V. See LTC4162-L datasheet Table 5 (p.4B) and DC2038A Table 1 (p.5)



Solar Powered 3-Cell 3.2A Charger with Maximum Power Point Tracking



Reference Design: https://www.analog.com/media/en/technical-documentation/eval-board-schematic/710-DC2038A_REV11_PCA_SCHEMATIC.pdf